MamaMind: A Voice-Enabled, Family-Centred Mobile Application for Maternal Wellness, Paternal Support, and Infant Care Coordination Using Firebase and Generative AI

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***Abstract*—Pregnancy and the postpartum period place heavy emotional, physical, and logistical demands on mothers, yet most digital tools address only one part of this period and treat the mother as the only user. This paper presents MamaMind, a family-centred mobile application that brings maternal wellness, paternal support, and infant care coordination into one cross-platform app. The mother, the father, and the baby each have a dedicated dashboard. A voice-first AI companion lets the mother speak a request, hear a spoken reply, or navigate the app by voice; a Dad Coach offers practical suggestions to the father; and shared tasks, check-ins, and encouragement messages synchronise through Firebase Authentication and Cloud Firestore. Baby-care features cover feeding, sleep, and diaper logs, milestones, vaccination reminders, and an age-based food chart, while self-care features include guided meditation, a mood journal, a Happy Jar, and stress-relief games. The app is built with Expo, React Native, and TypeScript, and its AI features are served by an Express backend that calls Google Gemini. MamaMind is presented as a working proof-of-concept; quantitative usability and performance evaluation is planned and is not reported here. It is a wellness and coordination aid, not a medical device.**

***Keywords*—maternal health, postpartum wellness, paternal support, mobile health, React Native, Firebase, generative AI, voice assistant, family care coordination**

I. Introduction

The months around childbirth are a period of intense change. Postpartum depression (PPD) is a major public health concern: a meta-analysis of 291 studies from 56 countries estimated a global pooled prevalence of 17.7% [1], and a systematic review of Indian studies estimated a pooled prevalence of 22% [2]. Fathers are affected as well. A meta-analysis of 43 studies found that about 10% of men experience depression between pregnancy and the first postpartum year, with higher rates reported three to six months after birth and a positive correlation between paternal and maternal depression [3]. Risk factors reported in Indian studies include financial difficulty, marital conflict, and lack of support from the husband [2]. At the same time, new parents must manage a dense daily routine of feeding, sleep, diaper changes, vaccinations, medication, and household work, usually while short of sleep.

Many parenting apps focus on a single function, such as logging baby activity, and assume a single user. Mental-health apps are usually designed for the individual rather than the household. Evidence suggests that apps can help: a meta-analysis of 176 randomized trials found small but significant effects of mental-health apps on depression and anxiety, with features such as mood monitoring and chatbot technology associated with larger effects [7]. MamaMind combines these ideas in one family-oriented app, so that the mother, the father, and the baby's records are connected rather than scattered across separate tools.

A. Problem Statement

New parents, and mothers in particular, need support that is available at any hour, usable with little effort, and shared with their partner. A practical solution should support the mother's emotional wellbeing without requiring her to type or navigate many screens, give the father concrete ways to help, keep baby-care records and reminders in one place, share tasks and status between family members in near real time, and run on commonly available smartphones using low-cost, widely available cloud services.

B. Objectives and Contributions

- A role-based mobile application with separate Mother, Father, and Baby dashboards, built as a single cross-platform codebase.
- A voice-first AI companion for the mother that accepts spoken requests, replies aloud, and can navigate the app by voice.
- A father-facing support layer comprising daily suggestions, a shared view of the mother's check-in, encouragement messages, a Dad Coach, and bonding ideas.
- A family space created or joined with an invite code, with shared tasks synchronised through Cloud Firestore.
- Integrated baby-care and self-care modules, including feeding, sleep, and diaper logs, vaccination reminders, meditation, a mood journal, and relaxation games.
- A low-cost proof-of-concept and foundation for a planned usability and performance evaluation with families.

II. Related Work

Digital support for the perinatal period can be grouped into screening and prevalence studies, app-based interventions, and conversational agents.

A. Maternal and Paternal Mental Health

PPD prevalence varies widely between countries; the global estimate of 17.7% ranged from 3% in Singapore to 38% in Chile [1]. The Edinburgh Postnatal Depression Scale (EPDS), a 10-item self-report questionnaire that takes about five minutes to complete, is the most widely used screening instrument and underpins many prevalence studies [1], [4]. Paternal depression is less studied but not rare, and it is associated with maternal depression [3]. These findings motivate an app that attends to both parents rather than the mother alone. MamaMind's wellness estimate is an app-level self-care aid and is not a substitute for a validated clinical instrument such as the EPDS.

B. Mobile and Digital Interventions

A systematic review of randomized trials reported that app-based psychosocial interventions improved EPDS scores and may help prevent PPD [6], while another review evaluated both the effectiveness and the quality of commercially available perinatal apps [5]. The evidence base is still mixed, which argues for treating apps as a complement to, not a replacement for, professional care. For general mindfulness apps, an updated meta-analysis of 45 trials found small but significant effects on symptoms of depression and anxiety relative to control groups [8]. These effects are modest, so MamaMind positions meditation, journaling, and games as low-effort daily self-care rather than treatment.

C. Conversational and Generative AI

A randomized trial of the Woebot conversational agent found it to be a feasible and engaging way to deliver cognitive behavioural self-help to young adults [9]. Large multimodal language models such as Gemini [11] now make it practical to build conversational features without training a custom model. In MamaMind, generative AI is used for supportive conversation and practical suggestions, such as Dad Coach ideas and packing checklists. It is not used to diagnose or to give clinical advice.

D. Research Gap

Existing work shows the potential of mental-health apps, conversational agents, and cloud-synchronised mobile software, but a practical gap remains for a single, low-cost app that serves the mother, the father, and the baby together; offers hands-free voice interaction; synchronises shared tasks and check-ins across family members; and bundles self-care, baby tracking, and reminders. MamaMind addresses this combination as a proof-of-concept.

III. System Architecture

MamaMind has four elements, shown in Fig. 1: (1) the mobile application used by the mother and the father, (2) an Express backend that exposes the AI features through Google GenAI, (3) Firebase Authentication and Cloud Firestore for identity and shared data, and (4) the shared family space that links the parents' accounts. The mobile application is the user-facing interface. It handles sign-up and login, role selection, the three dashboards, local reminders, and voice input and output.

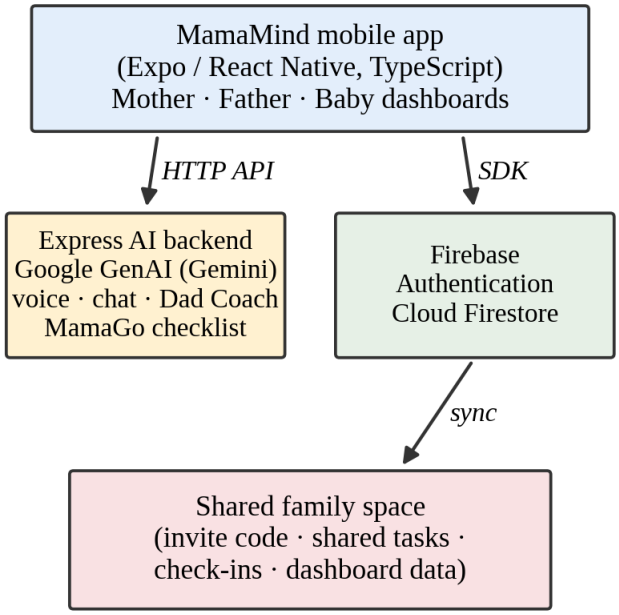


Fig. 1. MamaMind architecture: the mobile app calls the AI backend for generative features and Firebase for authentication and synchronised family data.

In the current design, requests that need generative AI, namely voice interaction, chat, Dad Coach, and MamaGo, are sent from the app to the backend, which calls Gemini and returns the result. Account and family data are read and written through the Firebase client libraries. When one parent adds a task or a check-in, the other parent's dashboard reflects it through Firestore synchronisation. Some preferences and logs are also kept locally with AsyncStorage.

The architecture follows five design requirements derived from the problem statement. Table I maps each requirement to the part of the system that addresses it.

TABLE I
Design Requirements and Responses

Requirement	How MamaMind responds
Low-effort interaction	Voice-first companion with spoken replies and voice navigation
Support for both parents	Separate Mother and Father dashboards with a shared check-in and encouragement messages
Records in one place	Baby dashboard with logs, milestones, vaccination schedule, and food chart
Near real-time sharing	Invite-code family space with tasks synchronised through Cloud Firestore
Low cost and wide reach	Single Expo / React Native codebase and managed cloud services

Separating the AI backend from the mobile client has two practical consequences. The client stays thin, and generative behaviour can be improved on the server. On the other hand, the AI features need a reachable backend, whereas logs, checklists, and reminders do not depend on it. The design therefore treats generative features as an enhancement on top of a core of ordinary, deterministic app functions.

Role-based access is also part of the architecture. A user chooses the Mother or Dad role at signup, and each role opens a different dashboard. Information is shared between the two

roles only through the family space: the father sees the mother's check-in only when she has added one, and shared tasks are visible to both. This keeps the mother in control of what she discloses to her partner, which matters because mood and wellness entries are personal.

IV. Technology Stack and Implementation Design

The app is implemented as a single TypeScript project built with Expo and React Native, which allows one codebase to target Android and iOS and, through React Native Web, a browser build [12], [13]. Navigation uses Expo Router, in which each screen is a file under the app directory. Table II lists the main components.

TABLE II
Technology Stack Components

Layer	Technology and purpose
Mobile framework	Expo SDK 57, React Native 0.86, React 19, and TypeScript; one codebase for the UI and logic
Navigation	Expo Router for screen routing between dashboards and feature screens
Identity and data	Firebase Authentication for signup and login; Cloud Firestore for family data and shared tasks
Local storage	AsyncStorage for data kept on the device
AI backend	Express 5 server calling Google GenAI (Gemini) for voice, chat, Dad Coach, and MamaGo
Device features	Notifications, image picker, audio, speech recognition, text-to-speech, and sensors via Expo modules
UI libraries	Reanimated, Gesture Handler, SVG, Safe Area Context, and Expo Vector Icons
Web support	React DOM and React Native Web

Cloud Firestore was chosen for data synchronisation because it is a serverless NoSQL database whose real-time notification system works with the Firebase client libraries to give mobile apps a smooth experience even across network connectivity problems [10], [14]. Generative features use Gemini, a family of multimodal models [11], through the Gemini API [15], which is called from an Express server [16] rather than directly from every screen. This keeps AI logic in one place, so prompts and models can be changed without republishing the app.

Several Expo modules provide the device capabilities the features rely on. Notification support schedules reminders for medication and vaccinations. The image picker lets parents attach photos to baby memories. Audio support plays the guided meditation and relaxation sounds. Speech recognition and text-to-speech provide the two ends of the voice companion, and the sensor module supports the wellness and wearable screens. Reanimated and Gesture Handler support the animated, touch-driven self-care games, and SVG is used for vector graphics.

The project is organised so that screens, services, and utilities are separate. Screens, such as the three dashboards and the feature pages, live in the app directory and are routed by Expo Router. Service modules wrap Firebase initialisation and authentication, and a utility module handles wearable data. The backend is a separate Node.js project with its own server file, so the mobile app and the AI service can be developed, configured, and deployed independently. This separation also makes it straightforward to replace the AI provider or to add new endpoints, such as a new assistant, without changing the dashboards.

V. Software and Data Design

A. Mobile Application

After login, the user selects a role (Mother or Dad) and is taken to the matching dashboard. Either parent can create a family space or join one using an invite code. The three dashboards are described below.

Mother dashboard. The dashboard centres on a voice-first AI companion. The mother can speak a request, hear the reply, or give voice commands to open screens such as baby care, family tasks, meditation, self-care, and community. A wellness check-in estimates wellbeing from logged activity or mood signals and shows a self-care recommendation. A medication screen stores medicine details and reminder times. MamaGo generates a personalised packing checklist for an outing with the baby, and items can be ticked off as they are packed.

Father dashboard. The father sees daily suggestions for supporting the mother, caring for the baby, or taking on household work. He can view the mother's check-in when she has shared one, send encouragement messages, and complete shared tasks; a support streak rewards consistency. Dad Coach provides practical AI-generated ideas, and a bonding section suggests activities for father and baby.

Baby dashboard. The baby profile stores the name, date of birth, and special-care notes. Daily feeding, sleep, and diaper logs show the last feed and last sleep, and a mood check-in records how the baby seems. Memories and milestones can include photos, and a supply checklist tracks essentials. A vaccination schedule provides reminders, and separate screens present growth and milestones and an age-based baby food chart.

B. Role-Based Feature Structure

Table III summarises the features by role. Self-care features are grouped under the mother's account, since they are aimed at her wellbeing, while family features apply to both parents.

TABLE III
MamaMind Feature Structure

Role	Features
Mother	Voice AI companion; wellness check-in; medication reminders; MamaGo checklist; guided meditation; mood journal; Happy Jar; self-care games; community posts; AI chat; wearable view
Father	Daily support suggestions; shared Mama check-in; encouragement messages; shared tasks and support streak; Dad Coach; bonding ideas
Baby	Profile; feeding, sleep, and diaper logs; mood check-in; memories and milestones with photos; supply checklist; vaccination schedule; growth screen; food chart
Family	Role selection, signup, and login; create or join by invite code; assign and complete shared tasks; Firestore synchronisation

The self-care games are Bubble Pop, Fidget Spinner, Pebble Stack, Zen Sand, Mood Garden, and Sound Garden. The daily care screen keeps a household task list with add, edit, and delete controls, and the Happy Jar lets the mother record positive moments to revisit later.

C. Voice Companion and AI Request Handling

Each voice interaction follows the procedure in Fig. 2. The app records the mother's speech and converts it to text with the device's speech recognition. If the text is a navigation command, the app opens the requested screen. Otherwise the text is sent to the backend, which uses Gemini to produce a supportive reply that the app speaks aloud with text-to-speech. Voice makes the companion usable when the mother's hands are occupied, for example while feeding the baby.

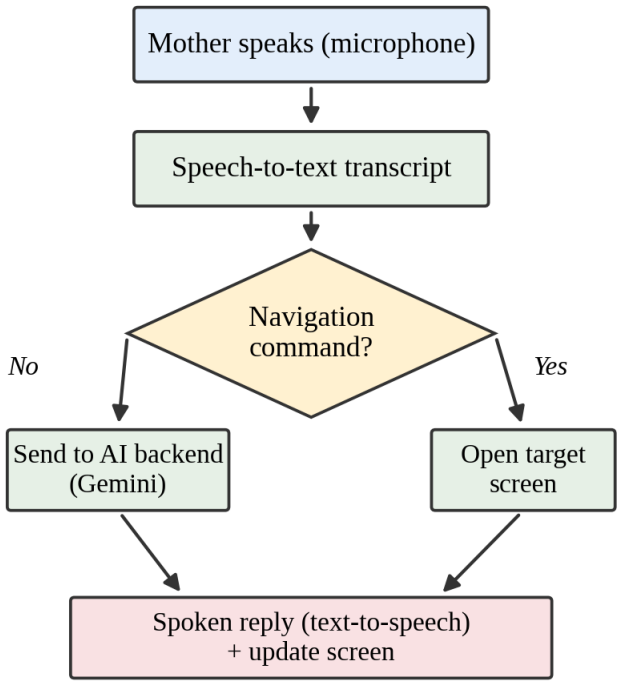


Fig. 2. Voice companion handling: speech recognition, navigation-command check, AI backend call, and spoken reply.

The same backend serves the text chat, Dad Coach, and MamaGo features, each with its own instructions. Dad Coach, for example, is prompted to return practical, specific actions for a father rather than general advice.

D. Family Synchronisation and End-to-End Workflow

The complete workflow is as follows. A parent signs up, chooses a role, and creates a family space; the invite code is shared with the partner, who joins from their own device. The mother logs her mood or wellness check-in and may share it. The father's dashboard shows the shared check-in and suggests ways to help; he sends an encouragement message or takes a shared task. Task changes are written to Firestore and appear on both devices. Throughout the day, either parent records feeding, sleep, and diaper events, and the app shows the latest values and sends reminders for medication and vaccinations. Voice requests and AI features run alongside this flow whenever the backend is reachable.

E. Reminders and Local Persistence

Time-based features are handled on the device. For medication, the mother enters the medicine details and one or more reminder times, and the app schedules local notifications. The vaccination feature uses the schedule built into the app to show what is due and to remind the parents. Baby logs, checklists, and some preferences are kept with AsyncStorage so that they remain available when the app is reopened. Because these functions do not call the AI backend, they continue to work when the backend is unavailable.

F. Responsible AI Considerations

Because the companion talks to a mother who may be tired, anxious, or low in mood, the AI features are intended to be supportive and practical rather than clinical. The responses are meant to offer comfort, simple coping suggestions, and ideas for next steps, and not to diagnose or prescribe. The wellness screen presents its output as a self-care recommendation, and simulated wearable values are labelled as such. These choices follow the evidence that app-based tools have modest effects and should complement professional care [5], [6], [8]. A production version would add explicit safety messaging, a path to professional or emergency help when a user expresses serious distress, and review of AI replies for harmful content.

VI. Experimental Setup

The MamaMind prototype is expected to be evaluated on Android and iOS devices through Expo, with the Express backend running on a development server and a dedicated Firebase project providing authentication and Firestore. A small group of volunteer participants, such as pairs of adults acting as mother and father roles, will be asked to carry out scripted tasks: creating and joining a family space, completing a wellness check-in, sending an encouragement message, assigning and completing a shared task, logging baby events, and issuing voice requests. Informed consent will be obtained, and no real clinical or infant health data will be collected.

For each task the evaluation will record completion, time taken, and observed errors. System measurements will include app start-up time, end-to-end voice response time, AI response time, shared-task synchronisation delay between two devices, and reminder delivery. A standard usability questionnaire will be administered after the session. Tests will be repeated on more than one device model and under different network conditions, so that results are not specific to a single phone or connection.

VII. Results and Discussion

The functional scope of the prototype was reviewed against the design goals in Table IV. The status reflects the features present in the current source code; on-device verification with the procedure in Section VI is still to be completed, so no pass or fail outcomes are claimed here.

TABLE IV
Functional Status

Function	Status
Role selection, signup, login	Implemented
Create / join family by invite code	Implemented
Shared task assignment and sync	Implemented
Voice companion (speech in, speech out)	Implemented
AI features: chat, Dad Coach, MamaGo	Implemented; needs backend
Feeding, sleep, diaper logs	Implemented
Vaccination schedule and reminders	Implemented
Wearable / wellness view	Implemented; values simulated

Quantitative performance must be reported from measurements on the built prototype rather than estimates. Table V lists the parameters to be measured.

TABLE V
QUANTITATIVE MEASUREMENTS

Parameter	Measured value
App start-up time	To be measured
Voice response time (end to end)	To be measured
AI response time	To be measured
Shared-task sync delay	To be measured
Reminder delivery success	To be measured
Task completion and usability score	To be measured

A. Advantages

MamaMind serves both parents and the baby in one app, so check-ins, tasks, and baby records are connected. Voice interaction lowers the effort of seeking support. The father is given specific actions rather than a general instruction to help, which addresses the lack of partner support reported as a risk factor for PPD [2]. A single TypeScript codebase reduces development effort across Android, iOS, and web [12], [13], and Firebase provides managed authentication and synchronisation without a custom database server [10], [14]. The bundled self-care tools are low-cost and can be used in short sessions between caring tasks.

B. Limitations

MamaMind is a wellness and coordination aid and not a medical device. The wellness screen labels its heart-rate and blood-oxygen values as simulated, so they must not be read as live medical readings. The wellness estimate is derived from logged activity and mood signals and has not been validated against an instrument such as the EPDS [4]. AI and family-synchronisation features depend on the backend and on Firebase being configured and reachable, and generative replies can be inaccurate, so they should not be treated as clinical advice. The app stores sensitive family information, which calls for careful security rules, privacy review, and consent before any real deployment. Because the evidence for app-based interventions is modest and mixed [5], [6], [8], the app should be used alongside, not instead of, professional care. The current work is a proof-of-concept and has not yet been tested with families.

C. Discussion

The prototype shows that the features of three separate app categories, namely baby tracking, maternal self-care, and partner coordination, can be delivered through one codebase with a modest technology footprint. Several design choices deserve comment. First, the role-based split means each parent sees only what is relevant to them, which keeps individual screens simple. Second, sharing through an invite-code family space is easy to explain, but it assumes two engaged adults; families with other structures, such as grandparents or other carers, would need a more flexible membership model. Third, the voice companion reduces effort for the mother but raises questions of privacy, since spoken content is processed by speech recognition and sent to a cloud AI service. A deployed system should tell users clearly what is sent where and let them choose text-only interaction.

The evidence reviewed in Section II suggests what to measure next. Because app-based interventions show small but real effects [7], [8], and because mood monitoring and conversational features are associated with larger effects [7], MamaMind's mood journal, wellness check-ins, and AI companion are the most promising components to study. The father-facing features address a gap that the literature highlights: paternal depression is common enough to matter and is linked to maternal depression [3], and lack of husband support is a reported risk factor in India [2]. A study comparing families who use the shared features with families who use only the mother's features would show whether partner involvement changes outcomes, such as perceived support or stress.

VIII. FUTURE WORK

- A formal usability and performance study with parents, following the plan in Section VI, including multiple device models and network conditions.
- Integration of validated screening, such as the EPDS [4], with clear escalation guidance to professional help.
- Real wearable data to replace the simulated heart-rate and blood-oxygen values.
- Offline-first behaviour for logs and reminders, with safe synchronisation when connectivity returns.
- Stronger privacy controls, including access rules, data export and deletion, and safety filters on AI replies.
- Regional-language voice and text support, and a clinician or community-health-worker view with parental consent.

IX. CONCLUSION

This paper presented MamaMind, a family-centred mobile application that combines maternal wellness, paternal support, and infant care coordination. A voice-first AI companion, a Dad Coach, shared tasks synchronised through Firebase, baby-care logs and vaccination reminders, and a set of self-care tools are delivered through role-based dashboards built with Expo and React Native, with generative features served by a Gemini-based Express backend. By treating postpartum care as a shared family effort, MamaMind provides a foundation for accessible, low-cost support. Usability and performance evaluation with families, validated screening, real wearable data, and stronger privacy safeguards will extend the system.

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